Côme Hosxe

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Available for full-time positions from November 2026

Education

Master Mathématiques, Vision, Apprentissage (MVA), ENS Paris-Saclay 2025 – 2026
- **Courses:** Deep Learning, Deep Generative Models, Graphical Models, Multimodal Explainable AI, Convex Optimization, Large-Scale Optimization, Deep Learning for Medical Imaging.

CentraleSupélec, Engineering degree (MSc), Université Paris-Saclay 2022 – 2026
- **Major:** Mathematics and Data Science. **GPA:** 4.33/4.33 (final year)
- **Courses:** Machine Learning, Reinforcement Learning, Computer Vision, Statistics, Stochastic Calculus, Harmonic Analysis.

Preparatory Class (MPSI, MP*), Lycée Louis-le-Grand, Paris 2020 – 2022

Professional Experience

Machine Learning Research Intern, Callyope, Paris May 2026 – Oct. 2026
- Research on reasoning in audio language models (ALMs) for speech-based mental-health assessment.
- Fine-tuned ALMs on self-generated chain-of-thought traces (STaR-style SFT, LoRA) and explored reinforcement learning with verifiable rewards (GRPO).
- Built an evaluation harness for open ALMs covering accuracy, calibration and audio grounding of reasoning, run at scale on HPC clusters (Jean Zay, Leonardo).

Machine Learning Research Intern, University of Auckland, New Zealand Mar. 2025 – Jul. 2025
- Designed and developed a deep learning model for bioacoustic classification, enabling real-time bird species identification from audio recordings.
- Developed a 3D sound localization algorithm based on MUSIC (Multiple Signal Classification).
- Built a streaming and recording daemon for field acoustic sensors (Bugg devices).

Data Scientist Intern, Datamaran, Valencia, Spain Jul. 2024 – Dec. 2024
- Developed a RAG system for extracting key information from corporate reports, exploring various embedding models, clustering techniques and deduplication methods.
- Studied cluster stability and implemented a web app to visualize and experiment with clustering.

Teaching Assistant, CentraleSupélec Sept. 2023 – Nov. 2023
- Taught lectures and tutorials to 30 first-year students: Bash, Git, OOP, SQL, networks and computer security.

Projects and Publications

Multi-Robot Multi-Drone Systems, Laboratoire des Signaux et Systèmes (L2S) Feb. 2023 – Jul. 2024
- Co-authored two published papers on educational games in Python integrating image processing, control algorithms and pattern recognition:
  - 2024 IFAC WACE: *Delivery Race Game: a Stimulative Approach to Engage Youths in Robotics and Control*
  - 2024 European Control Conference: *DroPong: Revisiting the Pong Game with Aerial and Ground Drones*

AI for Text-Based Deduction, CentraleSupélec Oct. 2025 – 2026
- Developed an AI agent playing a text-based deduction game ("Pedantix") using RL and NLP techniques.

President of Kampuchea Souriya, Volunteer Association Dec. 2022 – Dec. 2023
- Led a student solidarity association working in Cambodia, raised over $30,000 and built its website.

Skills

ML / LLM: PyTorch, Hugging Face Transformers, LoRA fine-tuning, vLLM, reinforcement learning (GRPO), Scikit-learn, OpenCV.

Engineering: Python, SQL, Bash, Git, SLURM, HPC.

Languages: French (native), English (C1), Spanish (B1), German (B1).

Hobbies: Freediving, hiking and outdoor activities, sudoku.